GCLK Kiosk Application Status and Requirements

Version 1.0 – 11/29/24

Introduction

This document outlines the current status and outstanding requirements for the GCLK Kiosk Application. While much progress has been made, additional refinements are needed to ensure full functionality and optimal performance. Further requirements may arise as the project enters its final stages.

Standard Open Application

The free application is currently deployed in the field and operational. However, the following issues or missing requirements have been identified:

1. **Attractor Screen Optimization**:
    - The current attractor screen is a looping GIF, which consumes significant system resources.
    - Proposed improvement: Develop a routine that allows the Portal to customize the attractor message using a simple text input. This text would be animated using vector-based transformations instead of pre-rendered GIFs, reducing resource usage. – OPTIONAL depends on effort required.
    - At a minimum the attractor screen GIF needs to be configurable on the Portal.
2. **Sheet Dispense Routine**:
    - The dispense routine needs to be revised for maximum efficiency and robustness to ensure accurate sheet dispense counts.
    - Exceptions to this accuracy should only occur due to external factors, such as sheets sticking together or other physical issues.
3. **Survey Questions (SQ) and Environmental Messages (EM)**:

    The sequence of survey questions and environmental messages must be configurable via the Portal. The following configurations are required:

    - **Config 1**: Standard survey question (SQ) followed by an environmental message (EM).
    - **Config 2**: Survey question 1 (SQ1) followed by survey question 2 (SQ2).
    - **Config 3**: Only survey question 1 (SQ1).
4. **Motor Control PCB Communication**:
    - The communication between the Raspberry Pi CM4 and the Motor Control PCB (MCB) via USB to serial must be reviewed and revised as needed.

- There is an intermittent issue with the connection to the MCB, which must be resolved. Collaboration with the engineer responsible for the MCB firmware is essential.

5. **Limit Switches for Inventory Tracking**:

   Limit switches are currently unused for tracking inventory levels. This functionality must be implemented as a redundant method to confirm inventory levels alongside the time-of-flight sensors.

Authenticated Application

The authenticated version of the application is partially complete, with some features (e.g., keyboard and login screen) already functional. The following tasks remain:

1. **Authentication Screen Completion**:

Finalize the authentication routine. Ensure that the format of the authentication (email address, phone number, or student ID) can be configured via the Portal.

2. **Requirements from the Open Application**:

All requirements listed for the Open Application also apply to the Authenticated Application.

Shared Application Requirements

Both the Open and Authenticated Applications must meet the following additional requirements:

1. **Portal Check-In**:

   Review and optimize the check-in process to ensure it efficiently provides the Portal with the necessary data.

2. **Usage Tracking**:

   Log all transactions and errors in a local database with date and time stamps for all entries.

3. **Credential Storage**:

   For the authenticated application, maintain a local database of all credentials specific to the kiosk or school.

4. **Software Updates**:

   Review the mechanism to update the application via a repository or similar method to ensure bug fixes and feature releases are automatically deployed. Current routine needs to be reviewed and approved or revised as needed.

5. **Software Watchdog**:

   Integrate a robust software watchdog to reset the program in the event of memory leaks or freezes, minimizing kiosk downtime.

6. **Version Display**:

   Display the current application version on the attractor screen.

7. **Sensor Reporting**:

   Send real-time sensor status to the Portal, including events such as door open, low inventory, and failed transactions.

8. **Transaction Data Reporting**:

   Send transaction data to the Portal after each dispense.

9. **Database Implementation**:

   Discuss database requirements with the development team to determine the most efficient implementation.

10. **Portal Configuration**:

    Ensure kiosk settings can be configured via the Portal as outlined in the Portal Requirements section.

Additional Scope Requirements

These features are outside the current project scope but should be discussed for future implementation:

1. **WiFi Configuration via Portal**:

   Enable administrators to configure kiosk WiFi settings remotely through the Portal.

2. **WiFi for RADIUS Servers**:

   Add support for specialized authentication methods, such as Eduroam, to connect to RADIUS servers.

Summary of Required Applications

1. User-Facing Applications

Two formats of the application will operate on the Raspberry Pi CM4 inside the kiosk:

1. **Open Application**:
    - No authentication required.
    - Tracks dispense transactions, survey questions and answers, and environmental messages displayed.
2. **Authenticated Application**:
    - Functions like the Open Application but includes an additional authentication screen for user credentials (email, phone number, or student ID). Credentials are logged in the local database.